



Blacktown Boys' High School

2018

HSC Trial Examination

Mathematics Extension 1

**General
Instructions**

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided for this paper
- All diagrams are not drawn to scale
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks** (pages 3 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7 – 11)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Assessor: X. Chirgwin

Student Name: _____

Teacher Name: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2018 Higher School Certificate Examination.

Section I

10 marks

Attempt Questions 1–10

Use the multiple choice answer sheet provided on page 13 for Questions 1–10.

1 Which of the following is an expression for $\int \sin^2 8x \, dx$?

A. $\frac{1}{2}x + \frac{1}{32}\sin 16x + C$

B. $\frac{1}{2}x - \frac{1}{32}\sin 16x + C$

C. $\frac{1}{2}x + \frac{1}{16}\sin 8x + C$

D. $\frac{1}{2}x - \frac{1}{16}\sin 8x + C$

2 Given that $f(x) = \frac{5}{x+2} + 3$.

The equations of the asymptotes of the graph of the inverse function $f^{-1}(x)$ are:

A. $x = -2$ and $y = 3$

B. $x = -2$ and $y = -3$

C. $x = 3$ and $y = -2$

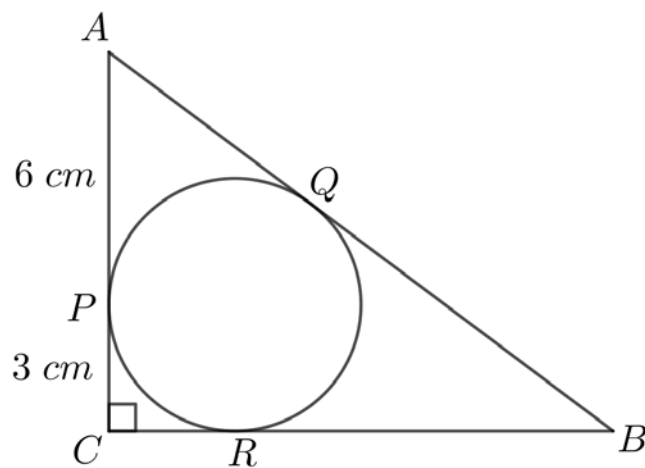
D. $x = -3$ and $y = -2$

- 3 Which of the following is the range of the function $y = \frac{1}{2} \cos^{-1} x - \frac{\pi}{2}$?
- A. $0 \leq y \leq \frac{\pi}{2}$
- B. $0 \leq y \leq \pi$
- C. $-\frac{\pi}{2} \leq y \leq 0$
- D. $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
- 4 How many arrangements in a row of the letters of the word AMPLITUDE are possible if all the vowels are together in any order?
- A. $6! \times 4!$
- B. $5! \times 4!$
- C. $9!$
- D. $4!$
- 5 In what ratio does the point $R(8, -1)$ divide the interval AB , where A and B are $(-4, 3)$ and $(5, 0)$ respectively?
- A. $-1 : 4$
- B. $-4 : 1$
- C. $4 : 1$
- D. $1 : 4$

- 6 The displacement x of a particle at time t is given by $x = 6 \sin 5t + 8 \cos 5t$. What is the greatest speed of the particle?

A. 70
 B. 50
 C. 30
 D. 10

- 7 In the diagram, AC is a tangent to the circle at the point P , AB is a tangent to the circle at the point Q , and BC is a tangent to the circle at the point R . Find the exact length of BC if $CP = 3 \text{ cm}$ and $AP = 6 \text{ cm}$.



A. 6 cm
 B. 9 cm
 C. 12 cm
 D. 15 cm

- 8 Which expression is equal to $\frac{{}^nP_r}{{}^nC_r}$?
- A. $\frac{n!}{(n-r)!}$
- B. $\frac{n!}{r!}$
- C. $\frac{1}{r!}$
- D. $r!$
- 9 Let $|p| \leq 1$, what is the general solution of $\cos 2x = p$?
- A. $x = n\pi \pm \frac{\cos^{-1} p}{2}, n \text{ is an integer}$
- B. $x = n\pi \pm \cos^{-1} \frac{p}{2}, n \text{ is an integer}$
- C. $x = 2n\pi \pm \cos^{-1} \frac{p}{2}, n \text{ is an integer}$
- D. $x = \frac{n\pi + (-1)^n \cos^{-1} p}{2}, n \text{ is an integer}$
- 10 The velocity $v \text{ ms}^{-1}$ of a particle moving in a straight line is governed by the equation $v = x - 3$, where x is its displacement. Initially, the particle was at $x = 7 \text{ cm}$. What is the displacement function of this particle?
- A. $x = 7 + e^t$
- B. $x = 7e^t$
- C. $x = 3 + e^t$
- D. $x = 3 + 4e^t$

End of Section I

Section II**60 Marks****Attempt Questions 11–14**

Answer each question on a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

a) Solve $\frac{4x}{x+5} < 2$. 2

b) Find the acute angle between the lines $x - 5y = 0$ and $2x + y - 3 = 0$. 2
Leave your answer to the nearest minute.

c) Find $\frac{d}{dx}(x \sin^{-1} x + \sin^{-1} x)$ 2

d) The variable point $(5 \cos \theta, 5 \sin \theta)$ lies on a curve. Find the Cartesian equation of this curve. 2

e) i) Prove that $\cot\left(\frac{\theta}{2}\right) = \frac{\sin \theta}{1 - \cos \theta}$. 2

ii) Hence find the exact value of $\cot\left(\frac{\theta}{2}\right)$ given that $\sin \theta = \frac{5}{6}$ and $\frac{\pi}{2} < \theta < \pi$. 2

f) Use the substitution $u = 2e^x$ to show that $\int_{\ln \frac{1}{2}}^{\ln \frac{\sqrt{3}}{2}} \frac{e^x}{1 + 4e^{2x}} dx = \frac{\pi}{24}$. 3

End of Questions 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- a) Consider the function $f(x) = \log_e(2x - 1) + x^3 + 1$.
- Show that a root exists between $x = 0.6$ and $x = 0.7$. 1
 - Use one application of Newton's method, starting at $x = 0.6$, to find another approximation of this root correct to 3 significant figures. 2
- b) The equation $3x^3 - 7x - 2 = 0$ has roots α , β and γ . Find the value of:
- $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 2
 - $\alpha^2 + \beta^2 + \gamma^2$ 2
- c) A Mathematics club consists of 20 members, of which there are 11 men and 9 women. A committee of four people is to be chosen randomly.
- How many committees can there be if there is to be equal numbers of men and women on the committee? 1
 - How many committees can there be if, regardless of their gender, the four-member committee must contain the eldest and youngest member of the club? 1
- d) Find an expression for the constant term in the expansion of $\frac{1}{x^2} \left(x^5 - \frac{3}{x} \right)^{16}$. 3
- e) Newton's law of cooling states that the rate of cooling of a body is proportional to the excess of the temperature of a body above the surrounding temperature.
 The temperature of a bowl of soup satisfies an equation of the form $T = B + Ae^{-kt}$ where T is the temperature of the soup, t is time in minutes, A and k are constants, and B is the temperature of the surroundings.
 The bowl of soup cools from 96°C to 88°C in 3 minutes in a room of temperature 25°C .
- Find the exact values of A and k . 2
 - Find the temperature of the bowl of soup, to the nearest degree, after a further 10 minutes have passed. 1

End of Questions 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

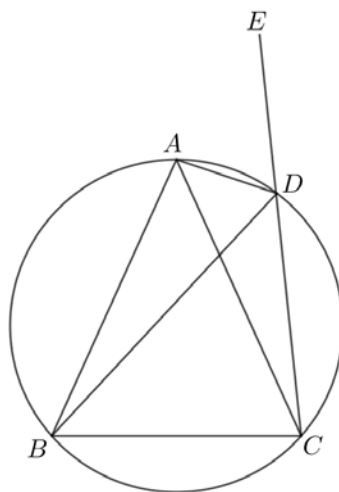
- a) The acceleration of a particle moving in a straight line is given by

$$\frac{d^2x}{dt^2} = -\frac{98}{x^2}$$

when x metres is the displacement from the origin after t seconds. When $t = 0$, the particle is 4 metres to the right of the origin with a velocity of 7 metres per second.

You may use the result $\frac{d^2x}{dt^2} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$.

- i) Show that the velocity, v , of the particle, in terms of x , is $v = \frac{14}{\sqrt{x}}$. 3
 - ii) Find an expression for t in terms of x . 2
 - iii) How many seconds does it take for the particle to reach a point 121 metres to the right of the origin? 1
 - iv) Find the displacement of the particle after 15 seconds. Round your answer to the nearest metre. 1
- b) $ABCD$ is a cyclic quadrilateral in which $AB = AC$, and CD is produced to E .



Not to
Scale

Copy or trace the diagram into your writing booklet.

- i) Explain why $\angle ABC = \angle ADE$. 1
- ii) Hence or otherwise, prove that AD bisects $\angle BDE$. 2

Question 13 continues on page 10

Question 13 (continued)

- c) i) Prove by mathematical induction that for all integers $n \geq 1$, **3**

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6}n(n+1)(2n+1).$$

- ii) Use this result to show that $2^2 + 4^2 + 6^2 + \cdots + 100^2 = 171700$. **1**

- iii) Hence evaluate $1^2 + 3^2 + 5^2 + \cdots + 99^2$. **1**

End of Questions 13

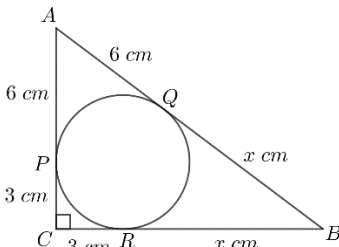
Question 14 (15 marks) Use a SEPARATE writing booklet.

- a) A man standing 100 metres from the base of a high-rise building observes an external lift moving up the outside wall of the building at a constant rate of 6.5 metres per second.
- i) If θ radians is the angle of elevation of the lift from the observer, 2
 show that $\frac{d\theta}{dt} = \frac{13 \cos^2 \theta}{200}$.
- ii) Evaluate $\frac{d\theta}{dt}$ at the instant when the lift is 40 metres above the observer's horizontal line of vision. 2
- b) The point $P(2ap, ap^2)$ lies on the parabola $x^2 = 4ay$. $S(0, a)$ is the focus of the parabola. The normal to the parabola at P meets the y -axis at R .
- i) Given that the equation of the normal at P is $x + py = ap^3 + 2ap$ (DO NOT PROVE THIS). Find the coordinates of R . 1
- ii) N lies on PR so that $SN \perp PR$. Find the coordinates of N . 2
- iii) Show that, as P moves along the parabola, the locus of N is another parabola. Find the vertex and focus of this parabola. 2
- c) A projectile is fired from the origin O with velocity V and with angle of elevation θ , where $\theta \neq \frac{\pi}{2}$. You may assume that
- $$x = Vt \cos \theta \quad \text{and} \quad y = -\frac{1}{2}gt^2 + Vt \sin \theta,$$
- where x and y are the horizontal and vertical displacements of the projectile in metres from O at time t seconds after firing.
- i) Show that the equation of flight of the projectile can be written as 2
 $y = x \tan \theta - \frac{1}{4h}x^2(1 + \tan^2 \theta)$, where $h = \frac{V^2}{2g}$.
- ii) Show that the point (X, Y) , where $X \neq 0$, can be hit by firing at two different angles θ_1 and θ_2 provided $X^2 < 4h(h - Y)$. 2
- iii) Show that no point above the x -axis can be hit by firing at two different angles θ_1 and θ_2 , satisfying $\theta_1 < \frac{\pi}{4}$ and $\theta_2 < \frac{\pi}{4}$. 2

End of Paper

2018 Mathematics Extension 1 Trial Solutions

Section 1

1	B $\int \sin^2 8x \, dx$ $= \int \frac{1}{2}(1 - \cos 16x) \, dx$ $= \frac{1}{2}\left(x - \frac{1}{16} \sin 16x\right) + C$ $= \frac{1}{2}x - \frac{1}{32} \sin 16x + C$	1 Mark
2	C $f(x) = \frac{5}{x+2} + 3$ Asymptotes of $f(x)$ are $x = -2$ and $y = 3$ Asymptotes of $f^{-1}(x)$ are $x = 3$ and $y = -2$	1 Mark
3	C $y = \frac{1}{2} \cos^{-1} x - \frac{\pi}{2}$ Range for: $y = \cos^{-1} x \rightarrow 0 \leq y \leq \pi$ $y = \frac{1}{2} \cos^{-1} x \rightarrow 0 \leq y \leq \frac{\pi}{2}$ $y = \frac{1}{2} \cos^{-1} x - \frac{\pi}{2} \rightarrow -\frac{\pi}{2} \leq y \leq 0$	1 Mark
4	A 4 vowels, A, I, U, E 4! ways of arranging all the vowels 1 group of vowels and 5 other letters 6! $\times$ 4!	1 Mark
5	B Let the ratio be $k:1$ $8 = \frac{5k + (-4) \times 1}{k + 1}$ $8k + 8 = 5k - 4$ $3k = -12$ $k = -4$ $\therefore -4 : 1$	1 Mark
6	B $x = 6 \sin 5t + 8 \cos 5t$ $\dot{x} = 30 \cos 5t - 40 \sin 5t$ Amplitude (greatest speed) = $\sqrt{30^2 + 40^2} = 50$	1 Mark
7	C Let $BR = x \text{ cm}$ $CP = CR = 3 \text{ cm}$; $AP = AQ = 6 \text{ cm}$; $BQ = BR = x \text{ cm}$ (two tangents from an external point have equal lengths) $AC^2 + BC^2 = AB^2$ (Pythagoras' Theorem) $(6 + 3)^2 + (3 + x)^2 = (6 + x)^2$ $81 + 9 + 6x + x^2 = 36 + 12x + x^2$ $6x = 54$ $x = 9$ $BR = 9 \text{ cm}$ $BC = 3 + 9 = 12 \text{ cm}$ 	1 Mark

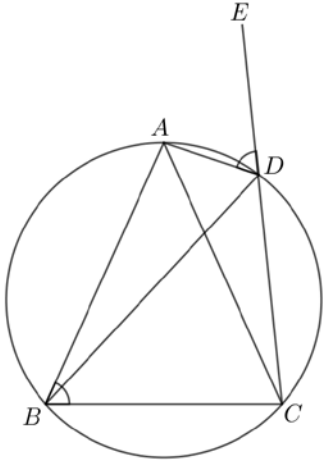
8	D $\frac{{}^nP_r}{{}^nC_r} = \frac{n!}{(n-r)!} \div \frac{n!}{(n-r)!r!}$ $\frac{{}^nP_r}{{}^nC_r} = \frac{n!}{(n-r)!} \times \frac{(n-r)!r!}{n!}$ $\frac{{}^nP_r}{{}^nC_r} = r!$	1 Mark
9	A $\cos 2x = p$ $2x = 2n\pi \pm \cos^{-1} p$ $x = n\pi \pm \frac{1}{2} \cos^{-1} p$	1 Mark
10	D When $t = 0, x = 7$ Only B or D are possible. In D, $x = 3 + 4e^t$ $v = \frac{dx}{dt} = 4e^t$ $4e^t = x - 3$ $v = x - 3$	1 Mark

Section 2		
Q11 a)	$\frac{4x}{x+5} < 2 \quad x \neq -5$ $4x(x+5) < 2(x+5)^2$ $4x(x+5) - 2(x+5)^2 < 0$ $(x+5)(4x - 2(x+5)) < 0$ $(x+5)(2x - 10) < 0$ $(x+5)(x - 5) < 0$ $-5 < x < 5$	2 Marks Correct solution 1 Mark Correctly identifies 5 and -5 as important, or equivalent merit
Q11 b)	$x - 5y = 0$ $y = \frac{1}{5}x$ $m_1 = \frac{1}{5}$ $2x + y - 3 = 0$ $y = -2x + 3$ $m_2 = -2$ $\tan \theta = \left \frac{\frac{1}{5} - (-2)}{1 + \frac{1}{5} \times (-2)} \right $ $\tan \theta = \frac{11}{3}$ $\theta = 74^\circ 44' 41.57''$ $\theta = 74^\circ 45' \text{ (nearest minute)}$	2 Marks Correct solution 1 Mark Correctly identifies the gradients of both lines and attempts to substitute into the formula
Q11 c)	$\frac{d}{dx}(x \sin^{-1} x + \sin^{-1} x)$ $= \frac{d}{dx}((x+1) \sin^{-1} x)$ $= \sin^{-1} x + \frac{x+1}{\sqrt{1-x^2}}$	2 Marks Correct solution 1 Mark Differentiate $\sin^{-1} x$ correctly
Q11 d)	$x = 5 \cos \theta$ $\cos \theta = \frac{x}{5}$ $y = 5 \sin \theta$ $\sin \theta = \frac{y}{5}$ $\sin^2 \theta + \cos^2 \theta = 1$ $\left(\frac{y}{5}\right)^2 + \left(\frac{x}{5}\right)^2 = 1$ $\frac{y^2}{25} + \frac{x^2}{25} = 1$ $x^2 + y^2 = 25$	2 Marks Correct solution 1 Mark Makes significant progress to eliminate θ
Q11 e) i)	Let $t = \tan \frac{\theta}{2}$ $\frac{\sin \theta}{1 - \cos \theta} = \frac{\frac{2t}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}}$ $\frac{\sin \theta}{1 - \cos \theta} = \frac{2t}{1+t^2 - (1-t^2)}$ $\frac{\sin \theta}{1 - \cos \theta} = \frac{2t}{2t^2}$ $\frac{\sin \theta}{1 - \cos \theta} = \frac{1}{t}$	2 Marks Correct solution 1 Mark Correct substitution of t formula

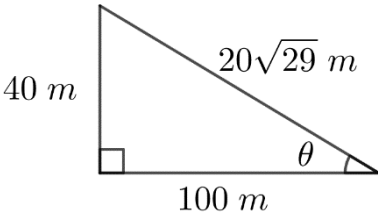
	$\frac{\sin \theta}{1 - \cos \theta} = \frac{1}{\tan\left(\frac{\theta}{2}\right)}$ $\frac{\sin \theta}{1 - \cos \theta} = \cot\left(\frac{\theta}{2}\right)$	
Q11 e) ii)	$\frac{\pi}{2} < \theta < \pi$ $\sin \theta = \frac{5}{6}$ $\cos \theta = -\frac{\sqrt{11}}{6}$ $\frac{\pi}{4} < \frac{\theta}{2} < \frac{\pi}{2}$ $\cot\left(\frac{\theta}{2}\right) = \frac{\sin \theta}{1 - \cos \theta}$ $\cot\left(\frac{\theta}{2}\right) = \frac{\frac{5}{6}}{1 - \left(-\frac{\sqrt{11}}{6}\right)}$ $\cot\left(\frac{\theta}{2}\right) = \frac{\frac{5}{6}}{\frac{6 + \sqrt{11}}{6}}$ $\cot\left(\frac{\theta}{2}\right) = \frac{5}{6 + \sqrt{11}}$	2 Marks Correct solution 1 Mark Find the correct exact value of $\cos \theta$
Q11 f)	$I = \int_{\ln \frac{1}{2}}^{\ln \frac{\sqrt{3}}{2}} \frac{e^x}{1 + 4e^{2x}} dx$ Let $u = 2e^x$ $du = 2e^x dx$ $x = \ln \frac{\sqrt{3}}{2}, u = \sqrt{3}$ $x = \ln \frac{1}{2}, u = 1$ $I = \frac{1}{2} \int_{\ln \frac{1}{2}}^{\ln \frac{\sqrt{3}}{2}} \frac{2e^x}{1 + (2e^x)^2} dx$ $I = \frac{1}{2} \int_1^{\sqrt{3}} \frac{du}{1 + u^2}$ $I = \frac{1}{2} [\tan^{-1}(u)]_1^{\sqrt{3}}$ $I = \frac{1}{2} [\tan^{-1} \sqrt{3} - \tan^{-1} 1]$ $I = \frac{1}{2} \left(\frac{\pi}{3} - \frac{\pi}{4} \right)$ $I = \frac{\pi}{24}$	3 Marks Correct solution 2 Marks Correct primitive function 1 Mark Obtains $du = 2e^x dx$ and correct boundary values in terms of u rather than x
Q12 a) i)	$f(x) = \log_e(2x - 1) + x^3 + 1$ $f(0.6) = \log_e(2 \times 0.6 - 1) + 0.6^3 + 1 = -0.3934 \dots$ $f(0.6) < 0$ $f(0.7) = \log_e(2 \times 0.7 - 1) + 0.7^3 + 1 = 0.4267 \dots$ $f(0.7) > 0$ Since there is a sign change, and the function is continuous for $0.6 \leq x \leq 0.7$, therefore there exist a root between 0.6 and 0.7.	1 Mark Correction solution

Q12 a) ii)	$f(x) = \log_e(2x - 1) + x^3 + 1$ $f(0.6) = \log_e(2 \times 0.6 - 1) + 0.6^3 + 1$ $f'(x) = \frac{2}{2x - 1} + 3x^2$ $f'(0.6) = \frac{2}{2 \times 0.6 - 1} + 3 \times 0.6^2$ $x = 0.6 - \frac{f(0.6)}{f'(0.6)}$ $x = 0.6 - \frac{\log_e(2 \times 0.6 - 1) + 0.6^3 + 1}{\frac{2}{2 \times 0.6 - 1} + 3 \times 0.6^2}$ $x = 0.6355 \dots$ $x = 0.636 \text{ (3 significant figures)}$	2 Marks Correct solution 1 Mark Correct differentiation of $f(x)$ and substitution of 0.6
Q12 b) i)	$3x^3 - 7x - 2 = 0$ $\alpha + \beta + \gamma = 0$ $\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{7}{3}$ $\alpha\beta\gamma = \frac{2}{3}$ $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$ $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{-\frac{7}{3}}{\frac{2}{3}}$ $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = -\frac{7}{2}$	2 Marks Correct solution 1 Mark Obtains correct sum and product of roots
Q12 b) ii)	$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $\alpha^2 + \beta^2 + \gamma^2 = 0 - 2 \times -\frac{7}{3}$ $\alpha^2 + \beta^2 + \gamma^2 = \frac{14}{3}$	2 Marks Correct solution 1 Mark Correct manipulation of the algebraic expression
Q12 c) i)	Choosing 2 men from 11 men and 2 women from 9 women ${}^{11}C_2 \times {}^9C_2 = 1980$	1 Mark Correct solution
Q12 c) ii)	2 member has already been set, so only choosing 2 people from the left over 18 people. ${}^{18}C_2 = 153$	1 Mark Correct solution
Q12 d)	$\left(x^5 - \frac{3}{x}\right)^{16} = \sum_{k=0}^{16} {}^{16}C_k (x^5)^{16-k} (-3x^{-1})^k$ $\left(x^5 - \frac{3}{x}\right)^{16} = \sum_{k=0}^{16} {}^{16}C_k x^{80-5k} (-3)^k x^{-k}$ $\left(x^5 - \frac{3}{x}\right)^{16} = \sum_{k=0}^{16} {}^{16}C_k (-3)^k x^{80-6k}$ Constant term in the expansion of: $\frac{1}{x^2} \left(x^5 - \frac{3}{x}\right)^{16} \rightarrow x^{-2} \times x^{80-6k} = x^0$	3 Marks Correct solution 2 Marks Attempts to solve k by matching up the correct terms in the expansion 1 Mark Obtains the correct expression for the binomial expansion

	$-6k = -78$ $k = 13$ Constant term is ${}^{16}C_{13}(-3)^{13}$	
Q12 e) i)	$T = B + Ae^{-kt}$ $t = 0, B = 25, T = 96$ $96 = 25 + Ae^0$ $A = 71$ $t = 3, B = 25, T = 88$ $88 = 25 + 71e^{-k \times 3}$ $\frac{63}{71} = e^{-3k}$ $\ln \frac{63}{71} = \ln e^{-3k}$ $-3k = \ln \frac{63}{71}$ $k = -\frac{1}{3} \ln \frac{63}{71}$	2 Marks Correct solution 1 Mark Obtains the correct value for A or k
Q12 e) ii)	Further 10 minutes $t = 3 + 10 = 13$ $T = 25 + 71e^{-\left(-\frac{1}{3} \ln \frac{63}{71}\right) \times 13}$ $T = 67.29 \dots$ $T = 67^\circ\text{C}$ (nearest degree)	1 Mark Correct solution
Q13 a) i)	$\frac{d^2x}{dt^2} = -\frac{98}{x^2} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$ $\frac{1}{2} v^2 = \int -98x^{-2} dx$ $\frac{1}{2} v^2 = \frac{-98x^{-1}}{-1} + C$ $v^2 = \frac{196}{x} + C$ $x = 4, v = 7$ $7^2 = \frac{196}{4} + C$ $C = 0$ $v^2 = \frac{196}{x}$ $\therefore v = \frac{14}{\sqrt{x}} \quad (v > 0)$	3 Marks Correct solution 2 Marks Correct integral for $\frac{1}{2} v^2$ and attempt to find C 1 Mark Express $\frac{1}{2} v^2 = \int -98x^{-2} dx$
Q13 a) ii)	$v = \frac{dx}{dt} = \frac{14}{\sqrt{x}}$ $\frac{dx}{dt} = \frac{14}{x^{\frac{1}{2}}}$ $\frac{dt}{dx} = \frac{x^{\frac{1}{2}}}{14}$ $t = \frac{1}{14} \int x^{\frac{1}{2}} dx$ $t = \frac{1}{14} \times \frac{x^{\frac{3}{2}}}{\frac{3}{2}}$ $t = \frac{1}{21} x^{\frac{3}{2}} + C$	2 Marks Correct solution 1 Mark Correct expression for t and attempt to find C

	$t = 0, x = 4$ $0 = \frac{1}{21} \times 4^{\frac{3}{2}} + C$ $C = -\frac{8}{21}$ $\therefore t = \frac{1}{21} x^{\frac{3}{2}} - \frac{8}{21}$	
Q13 a) iii)	$x = 121$ $t = \frac{1}{21} \times 121^{\frac{3}{2}} - \frac{8}{21}$ $t = 63 \text{ seconds}$ $\therefore$ It took 63 seconds for the particle to reach a point 121 metres to the right of the origin.	1 Mark Correct solution
Q13 a) iv)	$t = 15$ $15 = \frac{1}{21} x^{\frac{3}{2}} - \frac{8}{21}$ $15 + \frac{8}{21} = \frac{1}{21} x^{\frac{3}{2}}$ $x^{\frac{3}{2}} = \frac{323}{21} \div \frac{1}{21}$ $x^{\frac{3}{2}} = 323$ $x = 323^{\frac{2}{3}}$ $x = 47.07 \dots$ $x = 47 \text{ m (nearest metre)}$ $\therefore$ The particle is at 47 metres to the right of the origin after 15 seconds.	1 Mark Correct solution
Q13 b) i)	$\angle ABC = \angle ADE$ (exterior angle of a cyclic quadrilateral is equal to its opposite interior angle) 	1 Mark Correct solution
Q13 b) ii)	Let $\angle ABC = \theta$ $\triangle ABC$ is an isosceles triangle ($AB = AC$) $\angle ABC = \angle ACB = \theta$ (Equal base angles of isosceles $\triangle ABC$) $\angle ACB = \angle ADB = \theta$ (Angles in the same segment) $\angle ABC = \angle ADE$ (shown in the previous part) $\angle ADB = \angle ADE = \theta$ $\therefore AD$ bisects $\angle BDE$.	2 Marks Correct solution 1 Mark Identify $\angle ACB = \angle ADB$ and provided correct reasoning

Q13 c) i)	$1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1)$ 1. Prove statement is true for $n = 1$. $LHS = 1^2$ $LHS = 1$ $RHS = \frac{1}{6} \times 1 \times (1+1) \times (2 \times 1 + 1)$ $RHS = \frac{1}{6} \times 2 \times 3$ $RHS = 1$ $LHS = RHS$ $\therefore \text{Statement is true for } n = 1$ 2. Assume statement is true for $n = k$ (k some positive integer) $i.e. 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{1}{6}k(k+1)(2k+1)$ 3. Prove statement is true for $n = k + 1$ $i.e. 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{1}{6}(k+1)(k+2)(2k+3)$ $LHS = 1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2$ $LHS = \frac{1}{6}k(k+1)(2k+1) + (k+1)^2$ $LHS = \frac{1}{6}(k+1)[k(2k+1) + 6(k+1)]$ $LHS = \frac{1}{6}(k+1)[2k^2 + k + 6k + 6]$ $LHS = \frac{1}{6}(k+1)[2k^2 + 7k + 6]$ $LHS = \frac{1}{6}(k+1)(k+2)(2k+3)$ $LHS = RHS$ $\therefore \text{Statement is true by mathematical induction for all integers } n \geq 1.$	3 Marks Correct solution 2 Marks Makes significant progress in proving the statement involving $n = k + 1$ 1 Mark Establishes result for $n = 1$
Q13 c) ii)	$2^2 + 4^2 + 6^2 + \dots + 100^2$ $= 2^2(1^2 + 2^2 + 3^2 + \dots + 50^2)$ $= 4 \times \frac{1}{6} \times 50 \times (50+1)(2 \times 50 + 1)$ $= 171700$	1 Mark Correct solution
Q13 c) iii)	$1^2 + 3^2 + 5^2 + \dots + 99^2$ $= (1^2 + 2^2 + 3^2 + \dots + 99^2 + 100^2) - (2^2 + 4^2 + 6^2 + \dots + 100^2)$ $= \frac{1}{6} \times 100 \times (100+1) \times (2 \times 100 + 1) - 171700$ $= 166650$	1 Mark Correct solution
Q14 a) i)	$\tan \theta = \frac{h}{100}$ $\sec^2 \theta \, d\theta = \frac{1}{100} dh$ $\frac{d\theta}{dh} = \frac{1}{100 \sec^2 \theta}$ $\frac{d\theta}{dh} = \frac{\cos^2 \theta}{100}$	2 Marks Correct solution 1 Mark Obtains the correct expression of $\frac{d\theta}{dh}$ in terms of θ

	Or $\theta = \tan^{-1} \frac{h}{100}$ $\frac{d\theta}{dh} = \frac{1}{100^2 + h^2} \cdot 100$ $\frac{d\theta}{dh} = \frac{100}{100^2 + 100^2 \tan^2 \theta}$ $\frac{d\theta}{dh} = \frac{100 \sec^2 \theta}{\cos^2 \theta}$ $\frac{d\theta}{dh} = \frac{1}{100}$ $\frac{d\theta}{dt} = \frac{d\theta}{dh} \times \frac{dh}{dt}$ $\frac{d\theta}{dt} = \frac{100}{13 \cos^2 \theta} \times 6.5$ $\frac{d\theta}{dt} = \frac{200}{232}$	
Q14 a) ii)	When $h = 40$ $\tan \theta = \frac{40}{100}$ $\sqrt{40^2 + 100^2}$ $= \sqrt{11600}$ $= 20\sqrt{29}$ $\cos \theta = \frac{100}{20\sqrt{29}}$ $\cos^2 \theta = \frac{10000}{11600}$ $\cos^2 \theta = \frac{25}{29}$ $\frac{d\theta}{dt} = \frac{13 \cos^2 \theta}{200}$ $\frac{d\theta}{dt} = \frac{13}{200} \times \frac{25}{29}$ $\frac{d\theta}{dt} = \frac{13}{232} \text{ rad/s}$ 	2 Marks Correct solution 1 Mark Obtains the correct value for $\cos^2 \theta$
Q14 b) i)	At $R, x = 0$ $x + py = ap^3 + 2ap$ $py = ap^3 + 2ap$ $y = ap^2 + 2a$ $\therefore R(0, ap^2 + 2a)$	1 Mark Correct solution
Q14 b) ii)	$m_{SN} = m_T$ (gradient of the tangent at P) $x^2 = 4ay$ $y = \frac{x^2}{4a}$ $\frac{dy}{dx} = \frac{2x}{4a}$ $\frac{dy}{dx} = \frac{x}{2a}$ $m_T = \frac{2ap}{2a}$ $m_T = p$	2 Marks Correct solution 1 Mark Obtains the correct equation of SN

	Equation of SN $y - a = p(x - 0)$ $y = px + a$ Sub into $x + py = ap^3 + 2ap$ to find N $x + p(px + a) = ap^3 + 2ap$ $x + p^2x + ap = ap^3 + 2ap$ $x(1 + p^2) = ap^3 + ap$ $x = \frac{ap(p^2 + 1)}{p^2 + 1}$ $x = ap$ $y = p \times ap + a$ $y = ap^2 + a$ $\therefore N(ap, ap^2 + a)$	
Q14 b) iii)	$N(ap, ap^2 + a)$ $x = ap$ $p = \frac{x}{a}$ $y = ap^2 + a$ $y = a \times \left(\frac{x}{a}\right)^2 + a$ $y = \frac{x^2}{a} + a$ $ay = x^2 + a^2$ $x^2 = ay - a^2$ $x^2 = a(y - a)$ $\therefore$ Locus of N is another parabola with vertex $(0, a)$, the focal length is $\frac{a}{4}$, focus $\left(0, \frac{5a}{4}\right)$	2 Marks Correct solution 1 Mark Obtains the locus of N
Q14 c) i)	$x = Vt \cos \theta \dots \dots (1)$ $y = -\frac{1}{2}gt^2 + Vt \sin \theta \dots \dots (2)$ From (1) $t = \frac{x}{V \cos \theta}$ Substitute into (2) $y = -\frac{1}{2}g \left(\frac{x}{V \cos \theta}\right)^2 + V \left(\frac{x}{V \cos \theta}\right) \sin \theta$ $y = -\frac{gx^2}{2V^2 \cos^2 \theta} + \frac{x \sin \theta}{\cos \theta}$ $y = -\frac{gx^2 \sec^2 \theta}{2V^2} + x \tan \theta$ $y = -\frac{2gx^2 \sec^2 \theta}{4V^2} + x \tan \theta$ $y = \frac{2g}{V^2} \times -\frac{x^2 \sec^2 \theta}{4} + x \tan \theta$ $y = \frac{1}{h} \times -\frac{x^2 \sec^2 \theta}{4} + x \tan \theta \quad \left(\frac{V^2}{2g} = h\right)$ $y = -\frac{x^2(1 + \tan^2 \theta)}{4h} + x \tan \theta \quad (\sec^2 \theta = 1 + \tan^2 \theta)$ $y = x \tan \theta - \frac{1}{4h}x^2(1 + \tan^2 \theta)$	2 Marks Correct solution 1 Mark Substitute $t = \frac{x}{V \cos \theta}$ into y and attempts to simplify

Q14 c) ii)	$y = x \tan \theta - \frac{1}{4h} x^2 (1 + \tan^2 \theta)$ $4hy = 4hx \tan \theta - x^2 (1 + \tan^2 \theta)$ $4hy = 4hx \tan \theta - x^2 - x^2 \tan^2 \theta$ $x^2 \tan^2 \theta - 4hx \tan \theta + x^2 + 4hy = 0$ Substitute (X, Y) into the above: $X^2 \tan^2 \theta - 4hX \tan \theta + X^2 + 4hY = 0$ This quadratic equation in $\tan \theta$ has two distinct roots if $\Delta > 0$. $(-4hX)^2 - 4X^2(X^2 + 4hY) > 0$ $16h^2X^2 - 4X^4 - 16hX^2Y > 0$ $4h^2 - X^2 - 4hY > 0 \quad (\text{since } X^2 > 0)$ $4h^2 - 4hY > X^2$ $X^2 < 4h^2 - 4hY$ $X^2 < 4h(h - Y)$ If $X^2 < 4h(h - Y)$, there are two solutions, $\tan \theta_1$ and $\tan \theta_2$, for the equation, two different angles, θ_1 and θ_2 can be used to hit the point (X, Y).	2 Marks Correct solution 1 Mark Substitute (X, Y) and forms quadratic equation in $\tan \theta$
Q14 c) iii)	Let $\tan \theta_1, \tan \theta_2$ be roots of $X^2 \tan^2 \theta - 4hX \tan \theta + X^2 + 4hY = 0$ Product of roots $\tan \theta_1 \tan \theta_2 = \frac{X^2 + 4hY}{X^2}$ $\tan \theta_1 \tan \theta_2 = 1 + \frac{4hY}{X^2}$ $\tan \theta_1 \tan \theta_2 > 1 \quad (X^2 > 0, Y > 0)$ If both $0 < \theta_1 < \frac{\pi}{4}$ and $0 < \theta_2 < \frac{\pi}{4}$, then $0 < \tan \theta_1 < 1$ and $0 < \tan \theta_2 < 1$, so $\tan \theta_1 \tan \theta_2 < 1$. This contradicts to the product of roots. $\therefore$ No point above the x-axis can be hit from two different angles θ_1 and θ_2 satisfying $\theta_1 < \frac{\pi}{4}$ and $\theta_2 < \frac{\pi}{4}$.	2 Marks Correct solution 1 Mark Obtains the expression $\tan \theta_1 \tan \theta_2 > 1$ from the product of roots